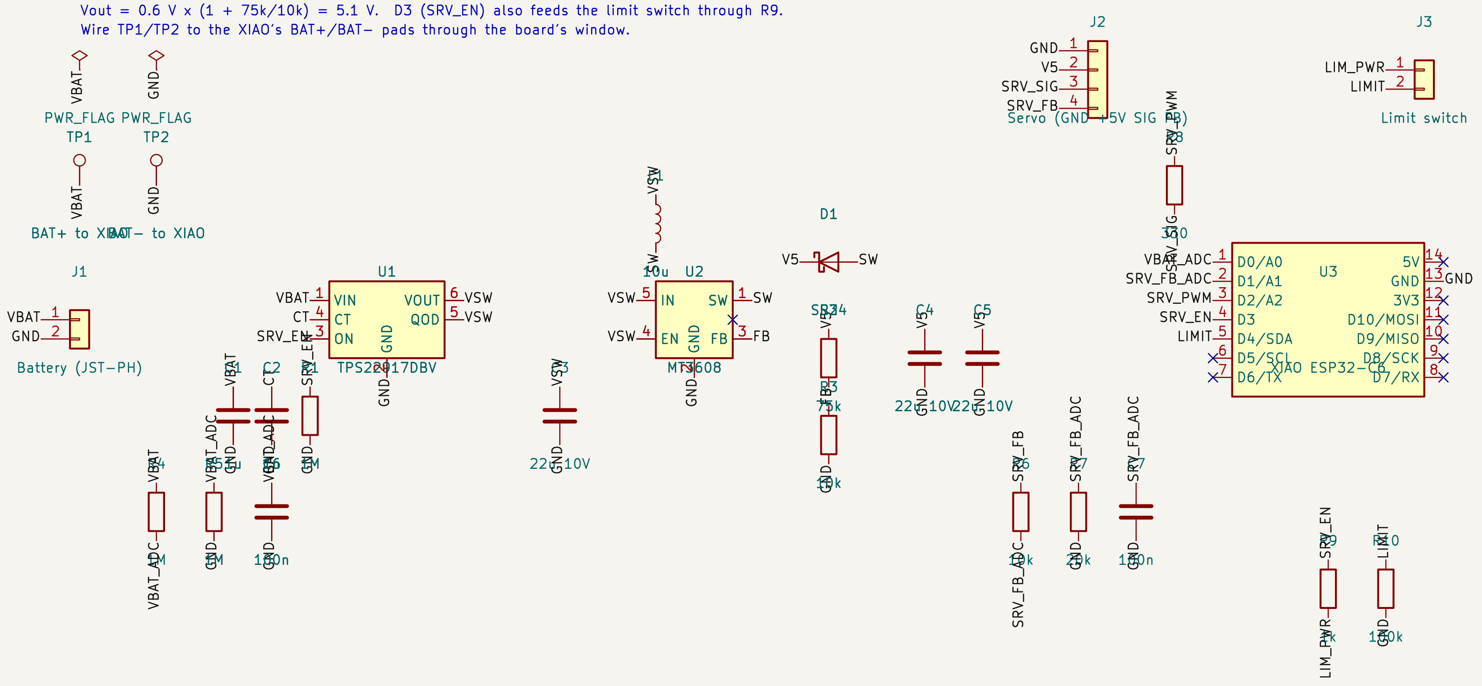
Smart vent backpack for Seeed XIAO ESP32-C6 (generated by generate.py)

Servo power: TPS22917 cuts the cell off completely (nA leakage), MT3608 boosts to 5.1 V.
Vout = 0.6 V x (1 + 75k/10k) = 5.1 V. D3 (SRV_EN) also feeds the limit switch through R9.
Wire TP1/TP2 to the XIAO's BAT+/BAT- pads through the board's window.



VBAT
GND
PWR_FLAG PWR_FLAG
TP1
TP2
BAT+ to XIAO
BAT- to XIAO
J1
Battery (JST-PH)
U1
VIN VOUT
CT QOD
ON GND
TPS22917DBV
VSW
SRV_EN
C1
C2
R1
R4
R5
C6
1M
100n
C3
22u 10V
L1
10u
U2
MT3608
IN SW
EN FB
D1
SS34
V5
SW
FB
R2
R3
75k
10k
C4
C5
22u 10V
J2
GND
V5
SRV_SIG
SRV_FB
Servo (GND +5V SIG FB)
R8
330
SRV_PWM
R6
R7
C7
10k
20k
SRV_FB_ADC
J3
LIM_PWR
LIMIT
Limit switch
U3
XIAO ESP32-C6
VBAT_ADC
SRV_FB_ADC
SRV_PWM
SRV_EN
LIMIT
D0/A0
D1/A1
D2/A2
D3
D4/SDA
D5/SCL
D6/TX
5V
GND
3V3
D10/MOSI
D9/MISO
D8/SCK
D7/RX
R9
R10
1k
100k

ben-hawks/yet-another-smart-vent

Sheet: /
File: smart_vent_backpack.kicad_sch

Title: Smart Vent XIAO Backpack

Size: A4 Date:
Rev: 0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1